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Alpilla 600 MW Combined Cycle Power Plant

Seawater Quality and Marine Conditions

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Function	Owner's Engineer (discipline lead)	Owner's Engineer (project engineer)	Owner's Project Director
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CASE STUDY - all organisations, data and signatures are fictional.

1. Basis

Twelve-month sampling campaign (September 2024 – August 2025) at three stations offshore of the Site (M1 at 300 m, M2 at 650 m, M3 at 1,000 m from the shoreline), monthly profiles at 1 m, 12 m and 30 m depth; continuous temperature logger at M2 (12 m). Bathymetry and current survey by the Owner's marine consultant (2025).

Important: the Sea of Marmara is a two-layer sea. The upper layer (Black Sea origin, brackish) lies above a sharp pycnocline at **20–25 m** depth; the lower layer (Mediterranean origin) is saline, colder in summer and low in dissolved oxygen, with occasional H₂S.

2. Bathymetry and hydrodynamics

- Seabed slope about 1:45; –12 m depth at 550 m, –15 m at 650 m (station M2), –30 m at 1,000 m from the shoreline.
- Tidal range negligible (< 0.1 m). Sea level variation from storm surge and barometric effects: –0.4 m to +0.6 m (100-year).
- Surface currents 0.1–0.4 m/s, predominantly westward; at 12 m 0.05–0.25 m/s.
- Waves: H_s 1-year 2.1 m; H_s 50-year 4.4 m (Tp 8.5 s), SW (Iodos) direction.

3. Seawater temperature at 12 m (station M2)

Month	Min (°C)	Mean (°C)	Max (°C)
Jan	7.8	9.0	10.4
Feb	6.9	8.1	9.3
Mar	7.4	8.6	10.1
Apr	9.6	11.2	13.5
May	13.1	15.4	18.2
Jun	17.0	19.8	23.1
Jul	20.4	23.2	26.4
Aug	21.6	24.1	27.0
Sep	19.8	22.0	24.6
Oct	15.9	18.1	20.7
Nov	12.6	14.3	16.8
Dec	9.8	11.0	12.9
Year	6.9	15.4	27.0

At 30 m (lower layer) the temperature is stable at 14.5–15.5 °C all year. Note: the draft contract uses 16.0 °C as Site Reference seawater temperature; the measured annual mean at 12 m is 15.4 °C.

4. Chemical quality (upper layer, 12 m; lower layer, 30 m for information)

Parameter	Unit	Upper layer min	Upper layer mean	Upper layer max	Lower layer mean
Salinity	PSU	21.8	23.6	26.1	38.2
Total dissolved solids	mg/l	22,100	24,000	26,600	39,000
Chloride	mg/l	12,000	13,100	14,500	21,200
Sodium	mg/l	6,700	7,300	8,100	11,800
Magnesium	mg/l	800	870	960	1,410
Calcium	mg/l	250	275	305	440
Potassium	mg/l	245	268	300	430
Sulphate	mg/l	1,700	1,830	2,020	2,960
Bicarbonate	mg/l	150	165	185	145
Boron	mg/l	2.4	2.9	3.4	4.5
Silica (SiO ₂)	mg/l	0.2	0.7	1.6	1.1
pH	-	7.9	8.15	8.35	7.8
Dissolved oxygen	mg/l	6.1	7.8	9.4	1.2 (occasionally 0)
H ₂ S	mg/l	n.d.	n.d.	n.d.	up to 0.4
Ammonium (NH ₄ -N)	mg/l	0.02	0.09	0.35	0.15
Iron (total)	mg/l	0.02	0.05	0.14	0.08
Manganese	mg/l	< 0.01	0.01	0.04	0.03

Parameter	Unit	Upper layer min	Upper layer mean	Upper layer max	Lower layer mean
TOC	mg/l	1.2	2.1	3.8	1.1
Oil and grease	mg/l	< 0.2	< 0.2	0.6	< 0.2
Total suspended solids	mg/l	3	9	48 (storm: 140)	5
Turbidity	NTU	0.8	3.5	22 (storm: 65)	1.5
SDI15 (raw)	-	4.8	> 5	not measurable	-
Chlorophyll-a	µg/l	0.5	2.8	14	0.3

n.d. = not detected (< 0.02 mg/l).

5. Biology and fouling

- Mussel (*Mytilus galloprovincialis*) settlement May–October, heavy.
- Jellyfish blooms (*Aurelia aurita*, *Mnemiopsis leidyi*): up to 20 kg/m³ of seawater locally during blooms (June–September).
- **Mucilage ("sea snot") events**: severe episodes in 2007 and 2021 (May–July) covering the surface and water column down to 25 m; the Owner requires the intake and screens to remain operable during such events.
- Fish spawning season (restriction on marine works per EIA): 1 May – 30 September.

6. Owner requirements arising

- Intake above the pycnocline, recommended depth –12 m, at least 3 m above seabed, velocity cap, intake velocity ≤ 0.15 m/s.
- Outfall at least 250 m from the intake, discharge in the upper layer.
- Materials: all seawater-wetted parts for chloride up to 14,500 mg/l and occasional H₂S.